Multi-Modal Deep Learning for Fashion Product Classification

Image + Text → product category

Team: A (image/CNN) · B (text/RNN) · C (data, fusion, infra)

Deep Learning course project

The question

> Does combining **image** and **text** beat using **either modality alone**?

- Input: a product **image** + its **text** description (`productDisplayName`)
- Output: the product's `subCategory` (top-10 classes)
- We train **three** models on the same pipeline and compare:
    i. 🖼️ image-only 2. 📝 text-only 3. 🔗 **fusion**

The 3-way comparison is the whole point of the project.

Dataset

- Kaggle **Fashion Product Images (small)** — ships images *and* labels *and* text
- Target: **`subCategory`**, top-10 most frequent classes
  - `masterCategory` → too easy (fusion invisible)
  - `articleType` → 140+ classes (too hard)
- Text field: `productDisplayName`
- **Stratified 70 / 15 / 15** split, fixed seed → identical for all 3 models

Architecture overview

```
image → EfficientNetB0 (frozen) → GAP → Dense(256) ─┐
                                                     ├→ Concat → Dense(256)
text  → TextVectorize → Embedding → GRU(128) → Dense(128) ─┘     → Dropout(0.3)
                                                                 → Softmax(10)
```

- Image-only & text-only baselines reuse the **identical encoders** → fair test
- Keras **functional API**, TensorFlow, runs on **Colab free GPU**

Image branch — CNN

- **Convolution**: shared local kernels → translation-equivariant features

$$Z_{i,j,f} = b_f + \sum_{c,u,v} K^{(f)}_{u,v,c}\, X_{is+u,\, js+v,\, c}$$

- **Activation** (nonlinearity): ReLU $\max(0, z)$; EfficientNet uses Swish $z\,\sigma(z)$
- **Pooling**: Global Average Pooling → one vector per channel (no params)
- **EfficientNetB0**, ImageNet weights, **frozen**: transfer learning, less overfit, cheap, BN in inference mode

Text branch — Embedding + GRU

- Trainable **embedding** $E \in \mathbb{R}^{V\times d}$ (no GloVe/BERT — short text)
- **GRU** gates control memory across the sequence:

$$z_t = \sigma(W_z e_t + U_z h_{t-1}), \quad r_t = \sigma(W_r e_t + U_r h_{t-1})$$

$$\tilde{h}_t = \tanh(W_h e_t + U_h(r_t \odot h_{t-1}))$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

- **Update gate** z_t = carry vs. write; **reset gate** r_t = forget history
- Final state h_L → Dense(128) = text encoding

Fusion — why concatenation + FC works

$$a = [\, a_{\text{img}}^{(256)} ; a_{\text{txt}}^{(128)} \,] \in \mathbb{R}^{384}$$

$$h_m = \mathrm{ReLU}\Big(\sum_i W_{m,i}^{\text{img}} a_{\text{img},i} + \sum_j W_{m,j}^{\text{txt}} a_{\text{txt},j} + b_m \Big)$$

- A hidden unit can fire only when **a visual feature *and* a textual feature co-occur** → learns **cross-modal conjunctions**
- When the image is ambiguous, the text disambiguates — and vice-versa
- Dropout(0.3) regularises the only high-capacity trainable part

Loss & optimisation

- **Softmax** → class probabilities; **cross-entropy** loss

$$\mathcal{L} = -\log \hat{y}_c, \qquad \frac{\partial \mathcal{L}}{\partial o_k} = \hat{y}_k - \mathbb{1}[k = c]$$

- **Adam** — adaptive per-parameter steps (momentum + RMS):

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

- Early stopping on val loss (patience 3, restore best); seed = 42 everywhere

Experimental setup

Optimiser	Adam, lr 1e-3
Loss	sparse categorical cross-entropy
Batch / epochs	32 / ≤15 (early stop)
Image	224×224, EfficientNetB0 frozen
Text	vocab 10k, seq len 20, emb 128, GRU 128
Eval	stratified held-out test (not k-fold — compute)
Tuning	small **manual** grid (lr, dropout, FC width)

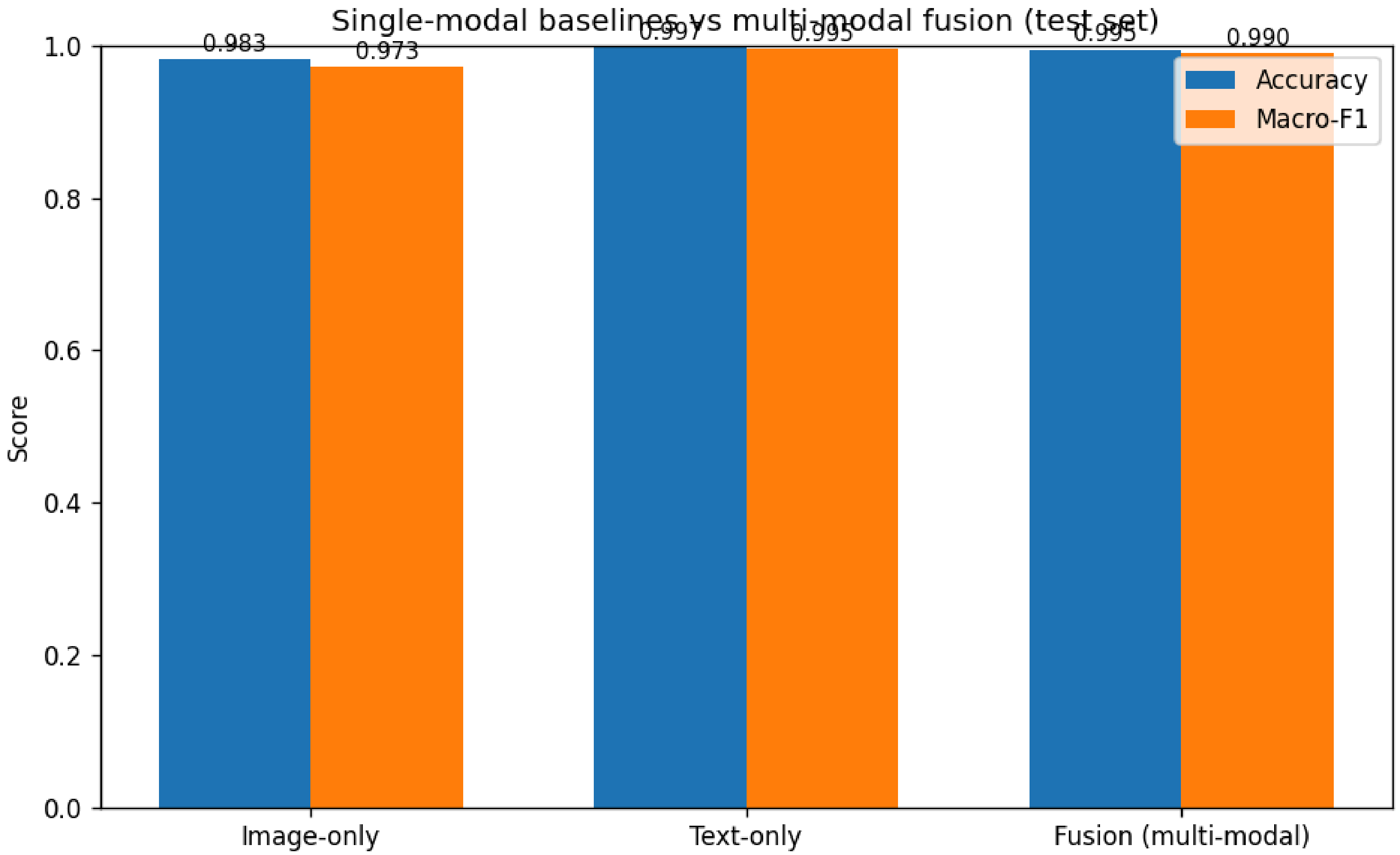
All hyperparameters centralised in `src/config.py`.

Results — the 3-way comparison

model	accuracy	macro-F1
Image-only	0.9827	0.9727
Text-only	**0.9970**	**0.9949**
Fusion	0.9947	0.9905

Headline: text **saturates** the task → fusion **ties** the best modality (ΔF1 −0.004, within noise) and both **beat image-only** (+0.018).

Results — visualised



Accuracy and macro-F1 are visually indistinguishable for text and fusion; image-only trails on both — fusion **matches** the dominant modality rather than beating it.

What the confusion matrices show

- Per-model `confusion_matrix.png` saved in `artifacts/<model>/`
- Fusion **repairs the image branch's worst confusions** by leaning on text:
    - 🖼️ image-only struggles on *Sandal* (F1 0.897, confused with *Shoes*) and *Innerwear* (0.941)
    - 🔗 fusion lifts them to **0.968** and **0.983**
- 📝 text-only is already near-perfect → fusion's gains show up **per-class**, not in the average
- We report **macro-F1** because the top-10 classes are imbalanced

Limitations & honest scope

- Small dataset images ~60×80 upscaled → visual ceiling
- `productDisplayName` is *very* informative → strong text baseline (less fusion headroom)
- Single held-out split (not k-fold) — justified by compute
- Backbone frozen (stage-2 fine-tuning = optional stretch)
- Trainable embedding instead of pretrained vectors

Takeaways

- Built the full mandated pipeline: **CNN + RNN + fusion**, end to end
- Derived the **math** of every component (conv/pool/activation, GRU gates, cross-entropy + Adam, fusion)
- **Fair** comparison: identical pipeline, shared encoders, no leakage
- **Reproducible**: fixed seed, persisted vocab, one-click Colab notebook
- Central result: text **saturates** this task → fusion **ties** it and both **beat image**; fusion's gains are **per-class** (recovers image's hardest categories)

Thank you — questions?